Answers

1. **Initial review and backup**
    - Create a copy of the raw data.
    - Convert the range to an Excel Table so formulas and analysis expand automatically with new data.
2. **Standardize inconsistent product names**
    - Remove extra spaces using `TRIM()`.
    - Fix casing using `UPPER()`, `LOWER()`, or `PROPER()`.
    - Correct spelling by creating a **reference (master) product list** and using `XLOOKUP()` or `VLOOKUP()` to map incorrect names to standard names.
3. **Split merged text columns**
    - Use **Text to Columns** (delimiter-based) if the separator is consistent.
    - Use formulas such as `LEFT()`, `MID()`, `RIGHT()`, `FIND()`, `TEXTBEFORE()`, `TEXTAFTER()` when patterns are fixed.
    - Store split fields (Region, Product, Code) as separate clean columns.
4. **Handle duplicate records without affecting totals**
    - Identify duplicates using **Conditional Formatting → Highlight Duplicates**.
    - Define a correct uniqueness rule (for example: Date + Product + Region + Invoice ID).
    - Instead of deleting blindly, use:
        - `COUNTIFS()` to flag true duplicates, or
        - Pivot Tables to aggregate correctly so totals remain accurate.
5. **Clean numeric fields with text and symbols**

- Remove symbols using `SUBSTITUTE()` (for example, currency signs or commas).
- Convert text to numbers using `VALUE()` or `--`.
- Use `ISNUMBER()` to validate numeric columns and flag invalid rows.

6. **Validate data**
   - Apply **Data Validation**:
     - Dropdown lists for Region and Product (from master lists).
     - Numeric limits for sales or quantity fields.
   - Add helper columns to mark records as “Valid” or “Invalid” using logical formulas.
7. **Filter and calculate required statistics**
   - Use `AVERAGEIFS()` and `STDEV.S()` with conditions to include only valid records.
   - Alternatively, use Pivot Tables with filters applied to Region, Product, and Valid flag.
8. **Final analysis and reporting**
   - Build Pivot Tables or summary formulas linked to the Excel Table.
   - Ensure all calculations reference structured table columns for dynamic updates.

Preventing future data issues

1. Use **Excel Tables** so new data is automatically included in cleaning rules and formulas.
2. Enforce **Data Validation rules** at the time of data entry.
3. Maintain **master reference tables** for products, regions, and codes.
4. Protect formula and helper columns to avoid accidental changes.

5. Standardize data entry templates and provide clear guidelines to all teams.

6. Periodically review flagged invalid or duplicate records.

Ensuring accuracy with new data

- Because formulas, validations, and Pivot Tables are built on Tables and reference lists, new rows automatically inherit:
    - Cleaning logic
    - Validation rules
    - Correct aggregation
- Refresh Pivot Tables to update summary statistics without manual rework.